

Study of the ketogenic agent AC-1202 in mild to moderate Alzheimer's disease: a randomized, double-blind, placebo-controlled, multicenter trial

Background Alzheimer's disease (AD) is characterized by early and region-specific declines in cerebral glucose metabolism. Ketone bodies are produced by the body during glucose deprivation and are metabolized by the brain. An oral ketogenic compound, AC-1202, was tested in subjects with probable AD to examine if ketosis could improve cognitive performance. Methods Daily administration of AC-1202 was evaluated in 152 subjects diagnosed with mild to moderate AD in a US-based, 90-day, randomized, double-blind, placebo-controlled, parallel-group study. Subjects were on a normal diet and continued taking approved AD medications. Primary cognitive end points were mean change from Baseline in the AD Assessment Scale-Cognitive subscale (ADAS-Cog), and global scores in the AD Cooperative Study – Clinical Global Impression of Change (ADCS-CGIC). AC-1202 was compared to Placebo in several population groups, including: intention-to-treat (ITT), per protocol, and dosage compliant groups. Results were also stratified by APOE4 carriage status (a predefined analysis based on the epsilon 4 (E4) variant of the apolipoprotein E gene). This trial was registered with ClinicalTrials.gov, registry number NCT00142805, information available at <http://clinicaltrials.gov/ct2/show/NCT00142805> Results AC-1202 significantly elevated a serum ketone body (β -hydroxybutyrate) 2 hours after administration when compared to Placebo. In each of the population groups, a significant difference was found between AC-1202 and Placebo in mean change from Baseline in ADAS-Cog score on Day 45: 1.9 point difference, $p = 0.0235$ in ITT; 2.53 point difference, $p = 0.0324$ in per protocol; 2.6 point difference, $p = 0.0215$ in dosage compliant. Among participants who did not carry the APOE4 allele (E4(-)), a significant difference was found between AC-1202 and Placebo in mean change from Baseline in ADAS-Cog score on Day 45 and Day 90. In the ITT population, E4(-) participants (N = 55) administered AC-1202 had a significant 4.77 point difference in mean change from Baseline in ADAS-Cog scores at Day 45 ($p = 0.0005$) and a 3.36 point difference at Day 90 ($p = 0.0148$) compared to Placebo. In the per protocol population, E4(-) participants receiving AC-1202 (N = 37) differed from placebo by 5.73 points at Day 45 ($p = 0.0027$) and by 4.39 points at Day 90 ($p = 0.0143$). In the dosage compliant population, E4(-) participants receiving AC-1202 differed from placebo by 6.26 points at Day 45 ($p = 0.0011$, N = 38) and 5.33 points at Day 90 ($p = 0.0063$, N = 35). Furthermore, a significant pharmacologic response was observed between serum β -hydroxybutyrate levels and change in ADAS-Cog scores in E4(-) subjects at Day 90 ($p = 0.008$). Adverse events occurred more frequently in AC-1202 subjects, were primarily restricted to the gastrointestinal system, and were mainly mild to moderate in severity and transient in nature. Conclusion AC-1202 rapidly elevated serum ketone bodies in AD patients and resulted in significant differences in ADAS-Cog scores compared to the Placebo. Effects were most notable in APOE4(-) subjects who were dosage compliant.